LABORATORY MANUAL

SC4172
IoT - TinyML

Lab Experiment #1

Training of Neural Network

SESSION 2025/2026
SEMESTER 2

COLLEGE OF COMPUTING AND DATA SCIENCE
NANYANG TECHNOLOGICAL UNIVERSITY

1. OBJECTIVES

Familiarize with the procedures to train a neural network using the TensorFlow/Keras machine learning library.

2. LABORATORY

This experiment is conducted at the **Software Laboratory 1 (Location: N4-01a-02)**.

3. HARDWARE EQUIPMENT

- Computer installed with Anaconda and Tools for TensorFlow ML.
- Arduino Nano 33 BLE Sense board (Optional for Lab #1)

4. REFERENCES

- TinyML book by Pete Warden, Daniel Situnayake. 2019. https://www.oreilly.com/library/view/tinyml/9781492052036/
- SC4172 Lecture Notes

5. INTRODUCTION

You will go through a sequence of tasks designed to improve your familiarity with using the Jupyter notebook to train a neural network to predict the output of a cosine function using a dataset artificially generated by the cosine function but added with noise as shown below.

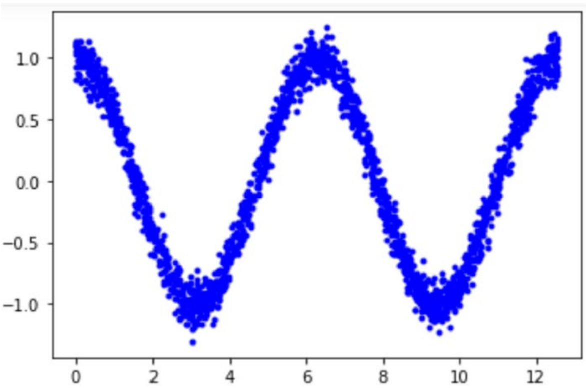


Fig. 1: Cosine Curve with Noise injection

The following are typical steps involved in training a neural network:

i. Decide on a goal
ii. Select, collect and label a dataset
iii. Design a model architecture
iv. Train the model using the training and validation dataset
v. Run the inference using the test data

6. PREPARATION

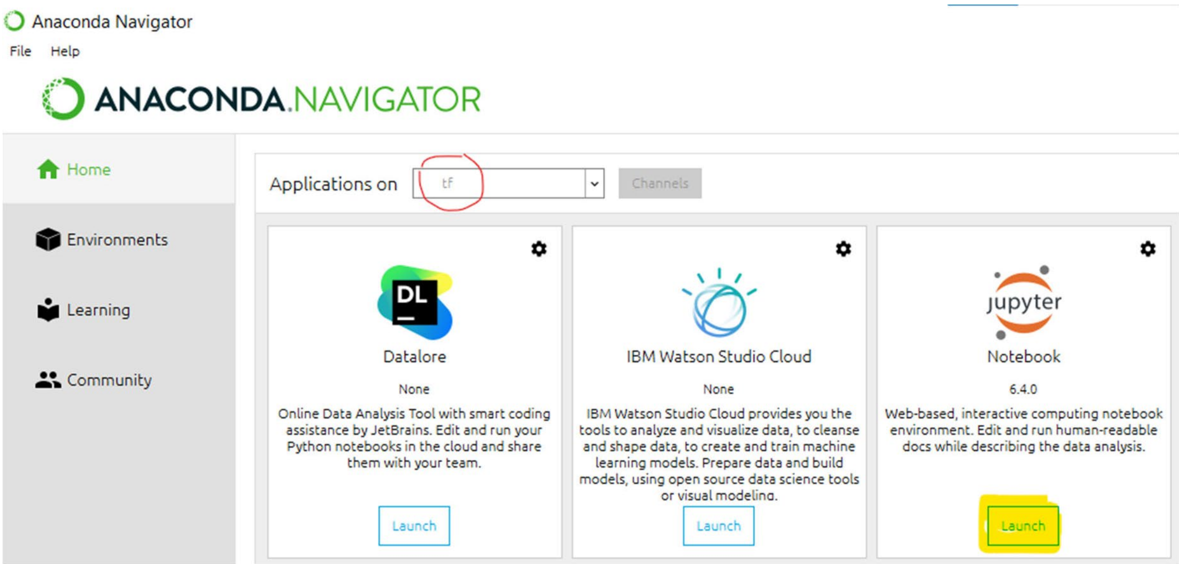
Turn on the PC and start the Anaconda Navigator application. The following

screenshots illustrate the first few steps use to initialize the environment for the

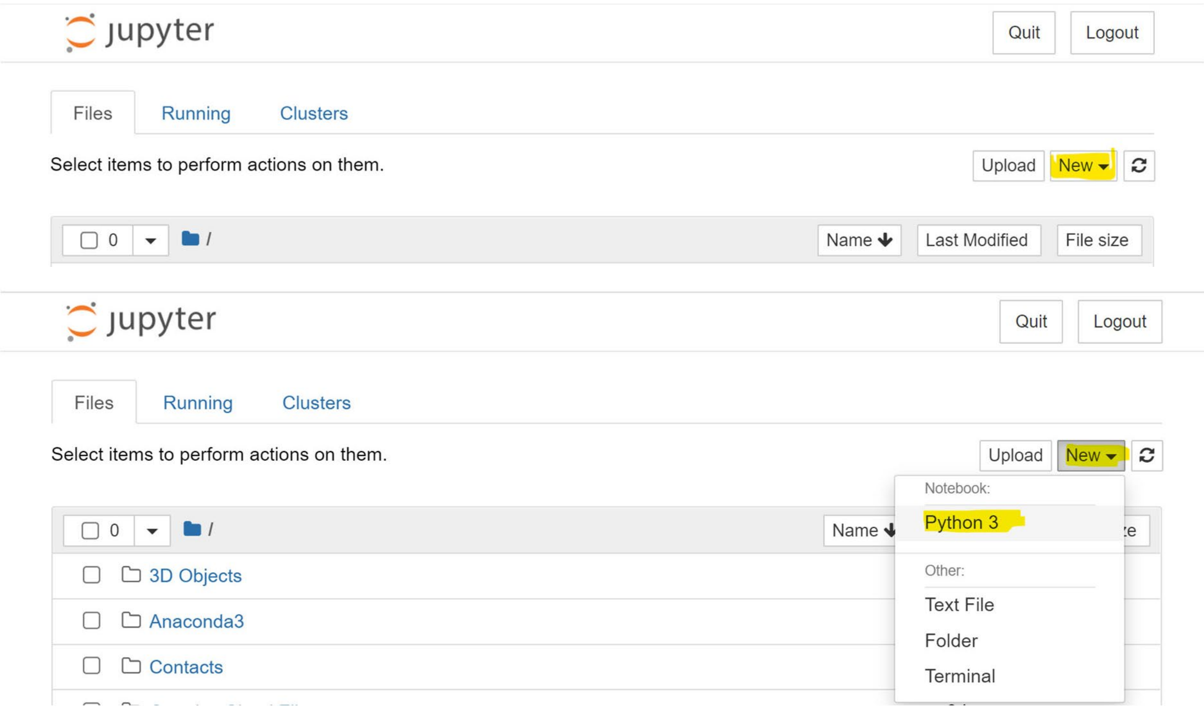
exercise. You should refer to the Lecture notes for the various instructions to be used in this exercise to verify and accomplish the goal.

i. Follow the steps outlined in the appendix to download and install Anaconda in the PC or Laptop.
   - Note that it is advisable to dothe installation on your laptop as you will need to use it for subsequent labs and your course project. It is alos ok to use the lab PC, just that you may need to restore any custome environment you have setup to ensure consistent behaviour in each lab (and your course project).
   - Note also that steps, behaviour and information outlined in the appendix may differ from what you might see due to difference in software versions. In which case, pleae adapt accordingly.

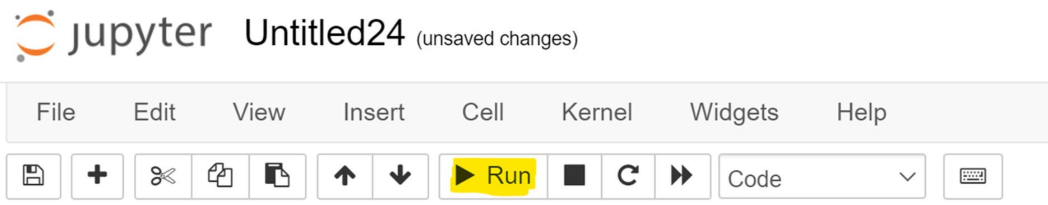
ii. Run Anaconda Navigator, select the 'tf' channel and launch the Jupyter Notebook application.



iii. Jupyter Notebook application will be opened in a web browser, select New and Python 3 to launch a new notebook.



iv. In the Notebook, key in the various instructions as shown below to generate a cosine dataset. Use the Run button to execute the instruction(s) in the cell..



In [1]:
```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
import math
```

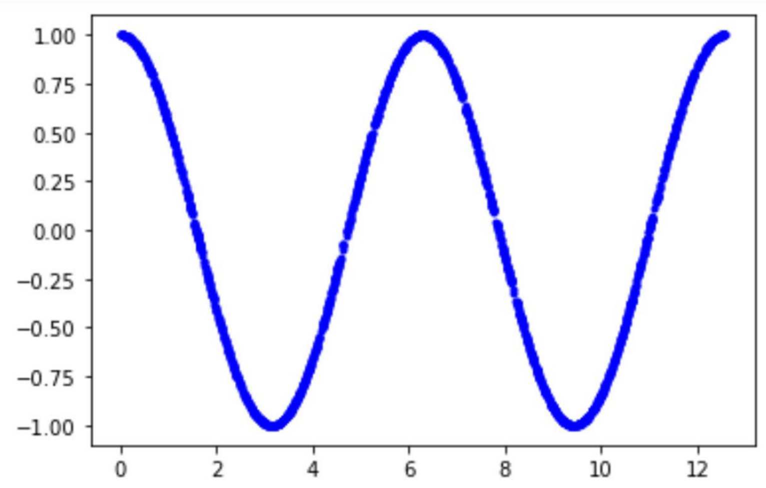
In [2]:
```
SAMPLES = 2000
SEED = 1337
np.random.seed(SEED)
tf.random.set_seed(SEED)
```

In [3]:
```
x_val=np.random.uniform(low=0, high=4*math.pi, size=SAMPLES)
```

In [4]:
```
np.random.shuffle(x_val)
```

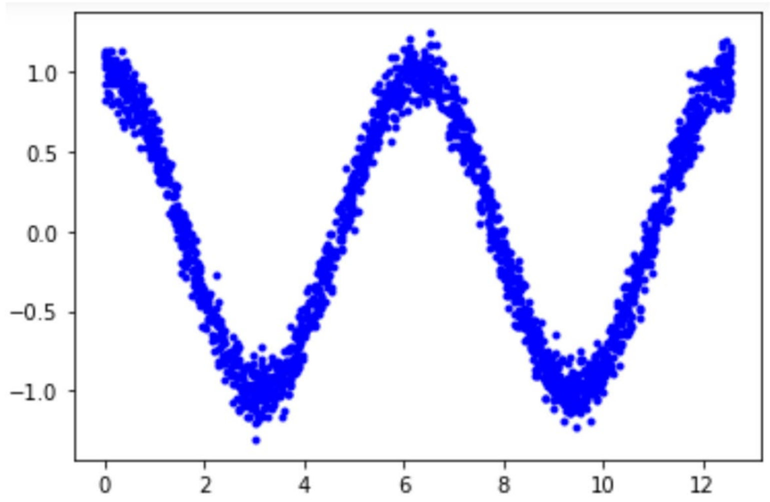
In [5]:
```
y_val=np.cos(x_val)
```

In [6]:
```
plt.plot(x_val, y_val, 'b.')
plt.show
```



7. GENERATING THE DATASET FOR TRAINING

Based on the dataset generated by the cosine function, add noise to the data to produce the 'Observed data' as shown below to be used to train the neural network. (Refer to the Lecture notes for the various instructions needed to accomplish this.)



8. DECIDING ON THE MODEL ARCHITECTURE

The next step is to decide on the model architecture to be used for the neural network. We would want to start with an architecture that is as simple and as small size as possible.

For practice, start with a single hidden layer (as shown in the Lecture) and using the 'ReLU' activation function.

9. TRAINING THE MODEL

Once the model architecture is decided, train the neural network. The step involves will be to first split the dataset into Training, Validation and Testing data. Refer to lecture note for the details involved in the training.

10. RUN INFERENCE AND CHECK THE PERFORMANCE

Once the training is completed, check its performance by using the Testing data. If the observed performance is not up to par, you will need to refine your model architecture and/or training approach by repeating step (iii) to step (v) in section 5 above.

11. COMPARE YOUR DESIGN AGAINST YOUR PEER

There are many possible ‘solutions’ to the same problem. You would want to compare your model architecture with those used by your classmate, such as the file size of the models and its run time.

APPENDIX

The following are steps that you can use to install the Anaconda Navigation and the various tools on your own computer/notebook for neural network training.

Tools required:

- Jupyter Notebook for Python programming
- TensorFlow and Keras for ML

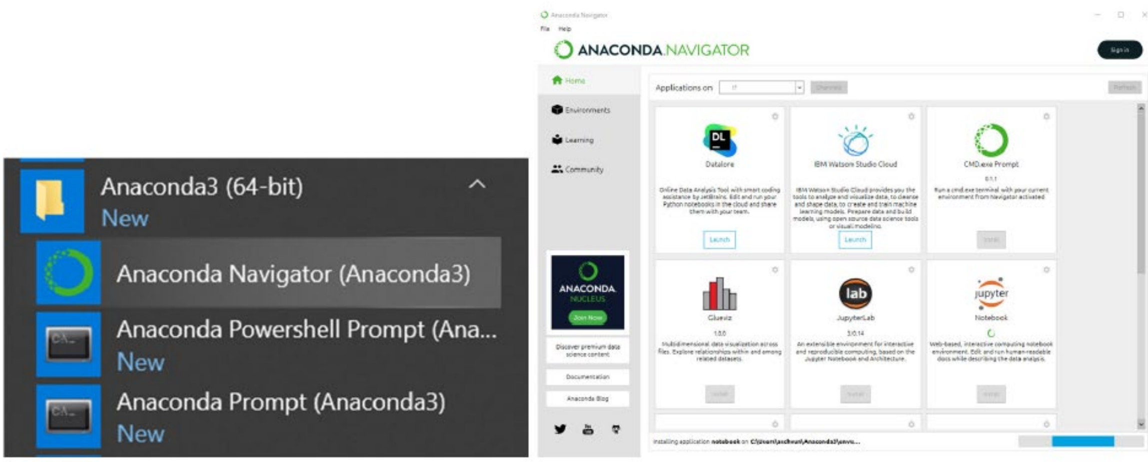
i. Download and install Anaconda

Visit the website https://anaconda.com/ and go to the "Products" and select "Individual".

Double click the installer and it will start installation.

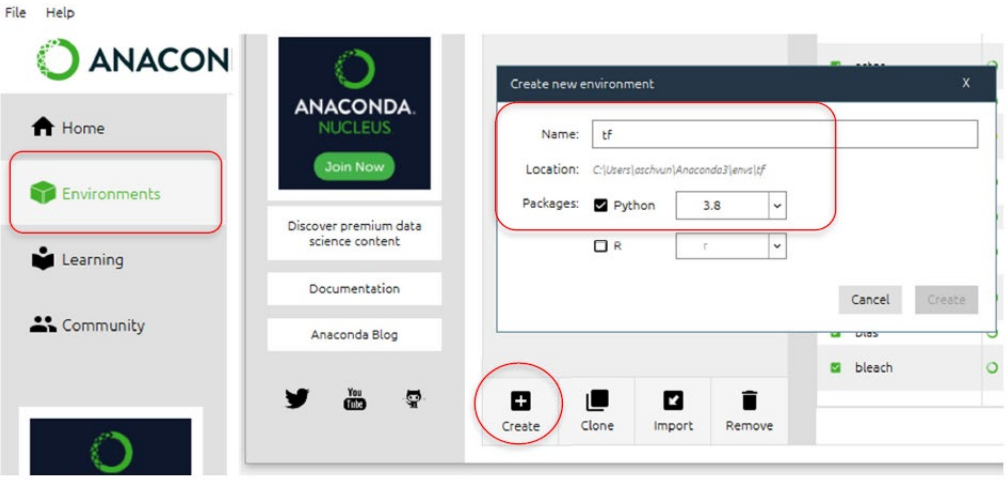
ii. Install TensorFlow with Anaconda

Run the Anaconda Navigator

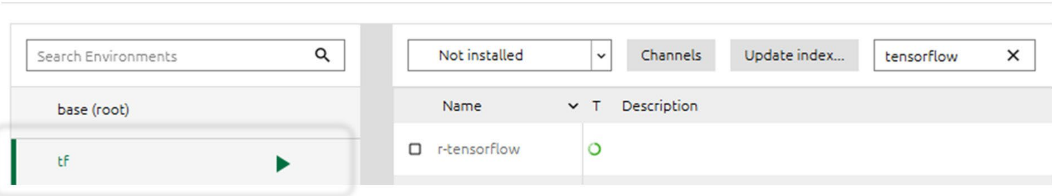


Click on the Environments tab, then Create.
Select 'Python' (and Python 3.8) for Packages and fill in 'tf' as the name of the environment

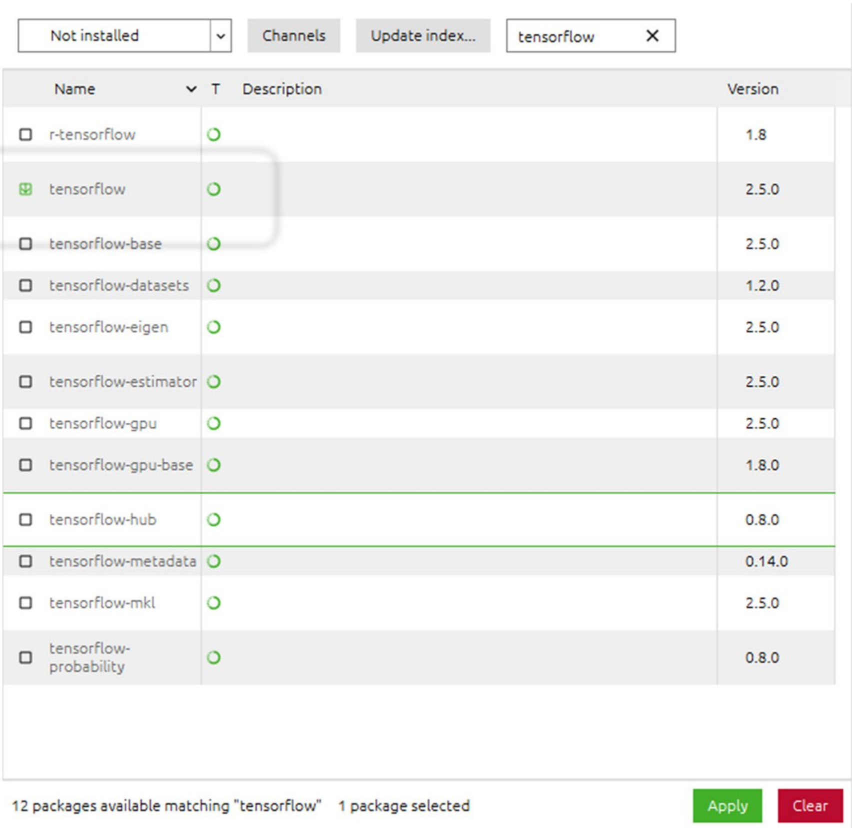
in the table



At the ‘tf’ environment, select ‘Not Installed’ and type in ‘tensorflow’

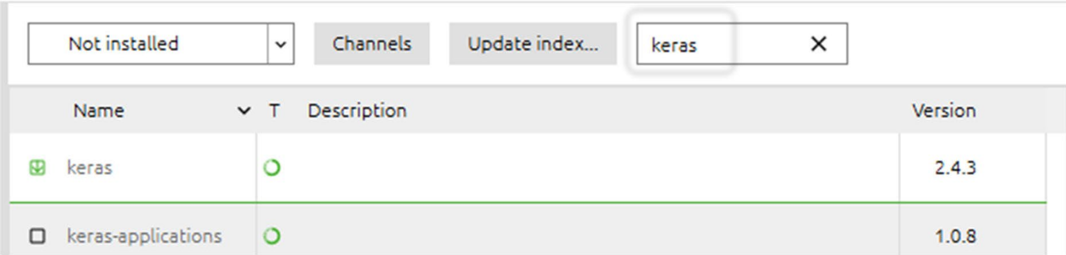


From the list, select ‘tensorflow’ click ‘Apply’ to install.

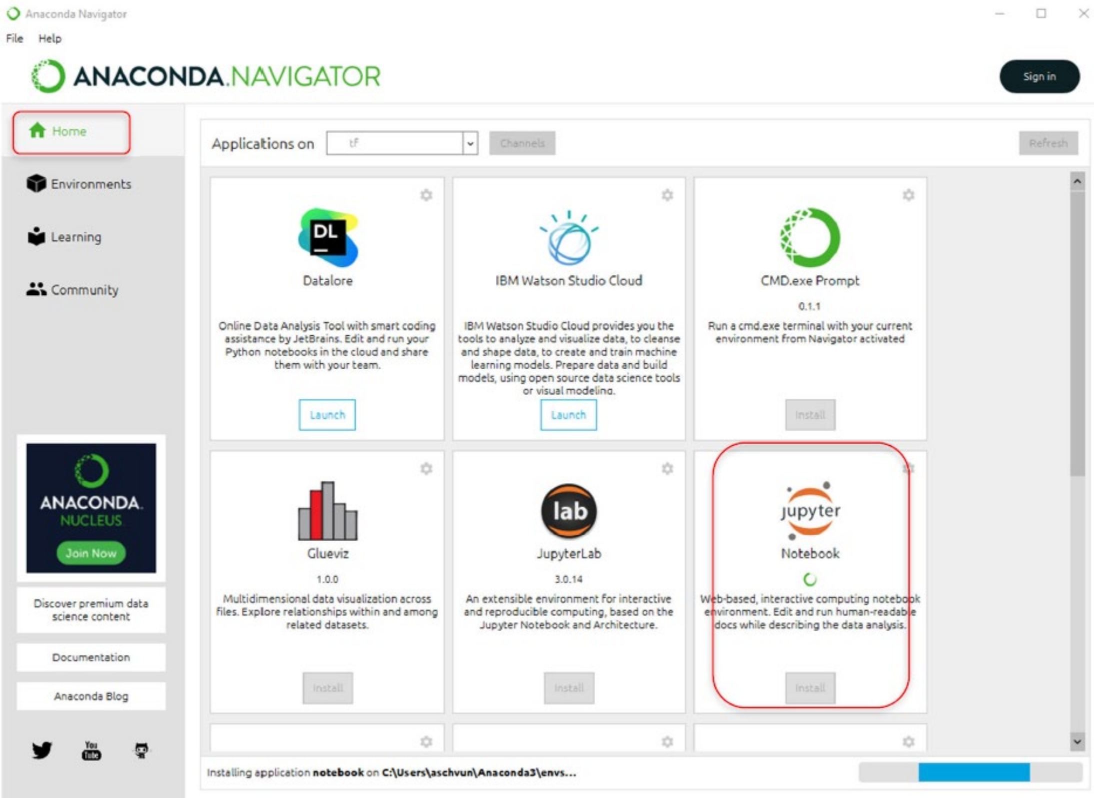


iii. Installation of other software package and tools

Use the same steps to install 'keras'

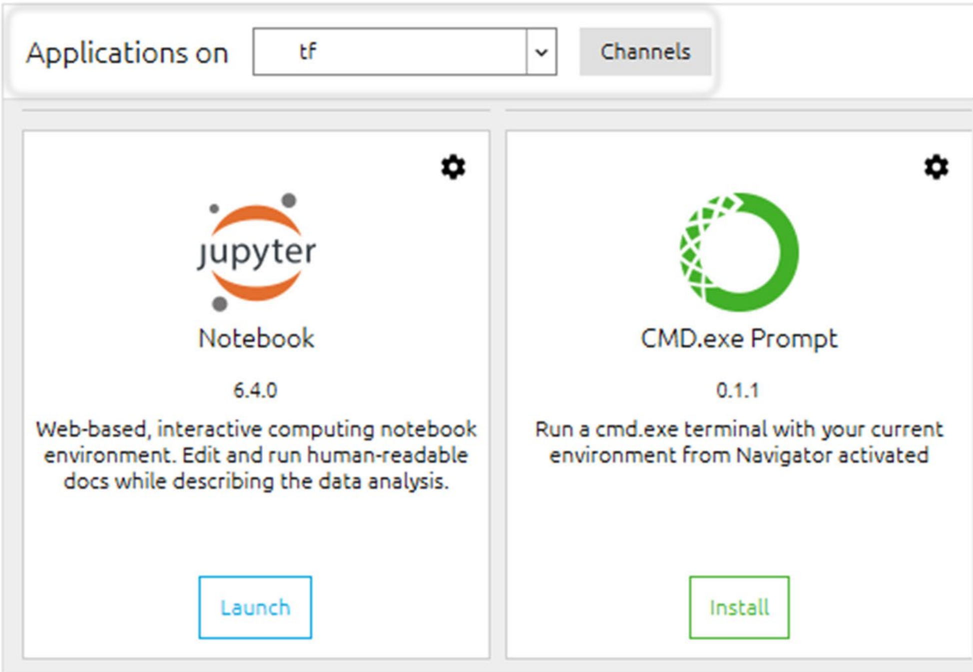


Back to 'Home', click to install "Jupyter Notebook"

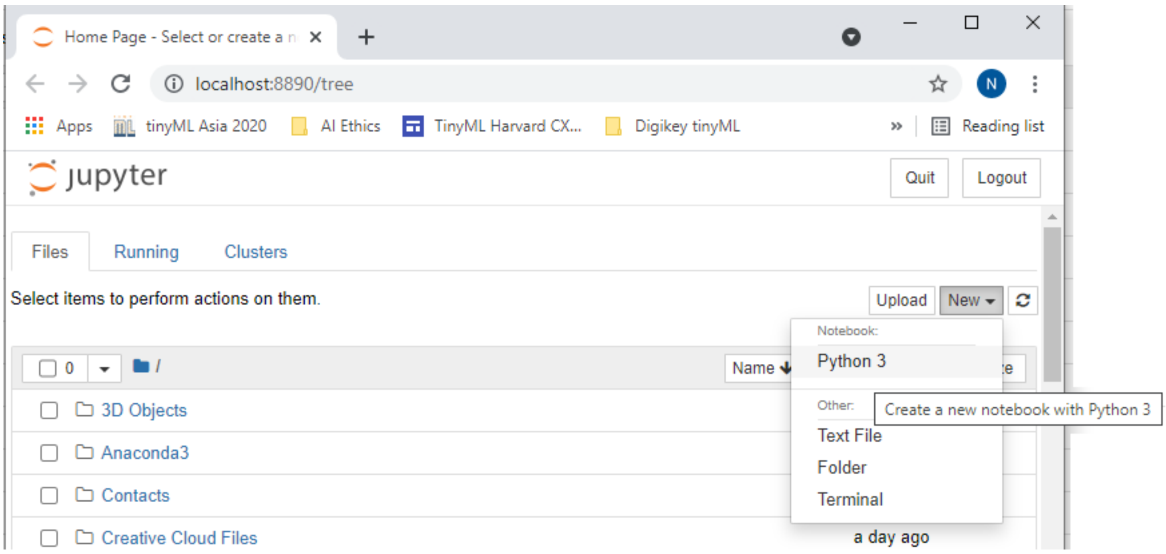


iv. Testing

Select the ‘tf’ Channel, then click to launch “Jupyter Notebook”

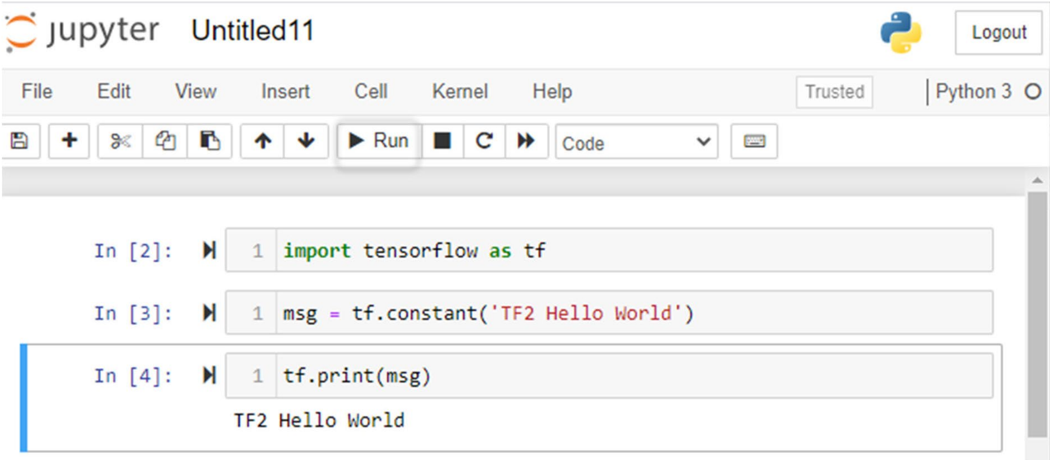


In “Jupyter Notebook”, select “New” and “Python 3”



Key in “import tensorflow as tf’ and then click ‘Run’, which should execute accordingly.

To further verify the setup, key in the message as shown below and see that the output is observed as shown.



Another library that you will use in developing ML programs which may need to be installed is matplotlib (only work with Python 3.8 as at 1/2022.)

Try the following, and if you get the error message as shown, then you should install the library ‘matplotlib’ through the Environment as shown earlier.